

DATABASE CAR PARKING SYSTEM

EXPERIMENT NO. 10

TITLE

Database with VIEW and INDEX

AIM

To create a VIEW and INDEX on the Car Parking System database.

DESCRIPTION

A VIEW is a virtual table based on a SELECT statement and provides a convenient way to access selected data. An INDEX improves data retrieval speed by helping MySQL locate matching rows efficiently.

QUESTIONS ON VIEW

1. Create a view named v1 using the VEHICLE table which displays vehicle number and owner name.

```
CREATE VIEW v1 AS
SELECT vehicle_no, owner_name
FROM VEHICLE;
```

```
SELECT * FROM v1;
```

2. Insert and delete records using a suitable updatable view.

```
CREATE VIEW v2 AS
SELECT vehicle_id, vehicle_no, owner_name
FROM VEHICLE;
```

```
INSERT INTO v2 VALUES (101, 'TN01AB1234', 'Arun');
```

```
DELETE FROM v2
WHERE vehicle_id = 101;
```

QUESTIONS ON INDEX

1. Create index1 for the amount attribute and display parking records whose amount is above 500. Explain indexing and drop index1.

```
CREATE INDEX index1 ON PARKING(amount);
```

```
SELECT vehicle_id, slot_id, amount
```

```
FROM PARKING
WHERE amount > 500;
```

```
DROP INDEX index1 ON PARKING;
```

2. Create index1 for vehicle_id and display the vehicle whose vehicle_id is 10. Then drop index1.

```
CREATE INDEX index1 ON VEHICLE(vehicle_id);
```

```
SELECT vehicle_no, owner_name
FROM VEHICLE
WHERE vehicle_id = 10;
```

```
DROP INDEX index1 ON VEHICLE;
```

WORKING PRINCIPLE OF INDEXING

An index stores searchable column values in an organized structure. When a query searches an indexed column, MySQL can locate matching rows faster than scanning the complete table.

OUTPUTS

1) View v1 displaying vehicle number and owner name.

vehicle_number	customer_name
TN10AA1111	Rahul
TN20BB2222	Meena
TN30CC3333	Suresh
TN40DD4444	Karthik
TN50EE5555	Sneha
TN60FF6666	Vijay
TN10AA1111	Rahul
TN20BB2222	Meena
TN30CC3333	Suresh
TN40DD4444	Karthik
TN50EE5555	Sneha
TN60FF6666	Vijay

12 rows in set (0.228 sec)

2) Record insertion and deletion through an updatable view.

customer_id	vehicle_number	customer_name
1	TN10AA1111	Rahul
2	TN20BB2222	Meena
3	TN30CC3333	Suresh
4	TN40DD4444	Karthik
5	TN50EE5555	Sneha
10	TN60FF6666	Vijay
11	TN70GG7777	Priya
101	TN10AA1111	Rahul
102	TN20BB2222	Meena
103	TN30CC3333	Suresh
104	TN40DD4444	Karthik
105	TN50EE5555	Sneha
106	TN60FF6666	Vijay

3) Parking records with amount above 500 using an index.

customer_id	vehicle_number	customer_name
1	TN10AA1111	Rahul
2	TN20BB2222	Meena
3	TN30CC3333	Suresh
4	TN40DD4444	Karthik
5	TN50EE5555	Sneha
10	TN60FF6666	Vijay
101	TN10AA1111	Rahul
102	TN20BB2222	Meena
103	TN30CC3333	Suresh
104	TN40DD4444	Karthik
105	TN50EE5555	Sneha
106	TN60FF6666	Vijay

4) Vehicle details for vehicle_id 10 using an index.

vehicle_number	customer_name	vehicle_type
TN60FF6666	Vijay	Car

RESULT

The records from the Car Parking System database were displayed using VIEW and INDEX.